Hull White Simulation 로직

1. 이론적 기초

HW 1Factor Process를 다음과 같이 가정한다.

$$\mathrm{r_t} = x_t + \phi_t$$

$$\mathrm{dx_t} = -\kappa x_t dt + \sigma_t dW$$

2. 확률과정 $\mathbf{e^{\kappa t} \cdot x_t}$에 대하여

$$\mathrm{d}(\mathrm{e}^{\kappa t} \cdot x_t) = \kappa e^{\kappa t} x dt + e^{\kappa t} dx_t = \kappa e^{\kappa t} x_t dt + e^{\kappa t}[-\kappa x_t dt + \sigma_t dW]$$

$$= \kappa e^{\kappa t} x_t dt + e^{\kappa t}(-\kappa x_t) dt + \sigma_t e^{\kappa t} dW = \sigma_t e^{\kappa t} dW$$

양 변을 T1에서 T2까지 적분하면

$$\mathrm{e}^{\kappa \mathrm{T}_2} \times x_{T_2} - \mathrm{e}^{\kappa \mathrm{T}_1} \times x_{T_1} = \int_{T1}^{T2} \sigma_t e^{\kappa t} dW$$

$$\mathrm{x_{T2}} = \frac{e^{\kappa T_1}}{e^{\kappa T_2}} x_{T_1} + \int_{T1}^{T2} \sigma_t e^{\kappa(t-T_2)}\, dW = e^{-\kappa(T_2-T_1)} x_{T_1} + \int_{T1}^{T2} \sigma_t e^{\kappa(t-T_2)}\, dW$$

$$r_{\mathrm{T}_2} = e^{-\kappa(T_2-T_1)} x_{T_1} + \int_{T1}^{T2} \sigma_t e^{\kappa(t-T_2)}\, dW + \phi_{T_2}$$

$$\mathrm{E}\left(\mathrm{x}_{\mathrm{T}_2} | T_1\right) = e^{-\kappa(T_2-T_1)} x_{T_1}$$

$$\mathrm{V}\left(\mathrm{x}_{\mathrm{T}_2} | T_1\right) = V\left(\int_{T_1}^{T_2} \sigma_t e^{\kappa(t-T_2)}\, dW | T_1\right) = \mathrm{E}\left(\left(\int_{T_1}^{T_2} \sigma_t e^{\kappa(t-T_2)}\, dW\right)^2 \middle| T_1\right)$$

$$= \mathrm{E}\left(\int_{T_1}^{T_2} \sigma_t^2 e^{2\kappa(t-T_2)}\, dt\right) = \int_{T_1}^{T_2} \sigma_t^2 e^{-2\kappa(T_2-t)}\, dt$$

3. 따라서, 차기 x(t)은 다음과 같이 계산

$$\mathrm{x(t)|x(s)} \sim N\left(e^{\kappa(t-s)} x(s), \int_s^t \sigma_u^2 e^{-2\kappa(t-u)} du\right)$$

$\mathrm{XA}(\mathrm{t_i}) = e^{-\kappa(t_{i+1}-t_i)}$ 이고, $\mathrm{XV}(\mathrm{t_i}) = \left(\int_{t_i}^{t_{i+1}} e^{-2\kappa(t_{i+1}-\tau)}\sigma^2(\tau)d\tau\right)^{0.5}$

만약 Vol이 Simple Vol이라면,

$$\mathrm{XV}(\mathrm{t_i}) = \left(\int_{t_i}^{t_{i+1}} e^{-2\kappa(t_{i+1}-\tau)}\sigma^2 d\tau\right)^{0.5} = \left(\frac{\sigma^2}{2\kappa}\left(1 - e^{-2\kappa(t_{i+1}-t_i)}\right)\right)^{0.5}$$

따라서, $\mathbf{x}(\mathbf{t_{i+1}}) = \mathbf{XA}(\mathbf{t_i}) \cdot \boldsymbol{x}(\boldsymbol{t_i}) + \boldsymbol{XV}(\boldsymbol{t_i}) \cdot \boldsymbol{\epsilon_i}$이며 다음과 같이 구현한다.

```
//A_k = exp(-kappa * (t_(k+1) - t_(k))
double XA(
    double kappa,
    double t0,
    double t1
)
{
    double A_k = exp(-kappa * (t1 - t0));
    return A_k;
}

double XV(
    double kappa,
    long NHWVol,
    double* HWVolTerm,
    double* HWVol,
    double t0,
    double t1
)
{
    long i;
    double vol = 0.0;
    double B_k, B_k_Square;
    kappa = max(0.00001, kappa);

    if (NHWVol == 1 || t1 - t0 < 0.25)
    {
        vol = Interpolate_Linear(HWVolTerm, HWVol, NHWVol, t0);
        B_k_Square = vol * vol * 0.5 / kappa * (1.0 - exp(-2.0 * kappa * (t1 - t0)));
    }
    else
    {
        long NInteg = 10.0;
        double u = t0;
        double du = (t1 - t0) / ((double)NInteg);
        B_k_Square = 0.0;
        for (i = 0; i < NInteg; i++)
        {
            vol = Interpolate_Linear(HWVolTerm, HWVol, NHWVol, u);
            B_k_Square += vol * vol * exp(-2.0 * kappa * (t1 - u)) * du;
            u = u + du;
        }
    }

    B_k = sqrt(B_k_Square);
    return B_k;
}
```

4. $\boldsymbol{P_{HW}(t,T) = E\left(e^{-\int_t^T r_u du}\middle| F_t\right) = E\left(e^{-\int_t^T (x_u+\phi_u)du}|F_t\right)}$**이다.**

x_t는 정규분포를 따르고, 위 수식은 정규분포 적률생성함수(MGF) 형태이다.

$$\mathbf{E}(\mathbf{e}^{\mathbf{tx}}) = \boldsymbol{e}^{\boldsymbol{\mu t}+\frac{\mathbf{1}}{\mathbf{2}}(\boldsymbol{\sigma t})^{\mathbf{2}}}$$

$$\mathbf{E}\left(-\int_s^t (\boldsymbol{x_u})\boldsymbol{du}\right) = \boldsymbol{B(s,t,\kappa)} = \int_s^t \boldsymbol{XA(u,t,\kappa)du}$$

$$= \int_s^t \boldsymbol{e}^{-\boldsymbol{\kappa}(\boldsymbol{t}-\boldsymbol{u})}\boldsymbol{du} = \frac{\mathbf{1}-\mathbf{e}^{-\boldsymbol{\kappa}(\mathbf{t}-\mathbf{s})}}{\boldsymbol{\kappa}}$$

$$\frac{\mathbf{1}}{\mathbf{2}}\mathbf{V}\left(-\int_s^t \boldsymbol{x_u du}\right) = \frac{\mathbf{1}}{\mathbf{2}}\mathbf{VT}(\mathbf{T_1},\mathbf{T_2},\boldsymbol{\kappa},\{\boldsymbol{\sigma}\})$$

$$\approx \frac{\sigma^2}{2\kappa^2}\left(\mathrm{T}-\mathrm{t}+2\frac{\mathrm{e}^{-\kappa(\mathrm{T}-\mathrm{t})}-1}{\kappa}-\frac{\mathrm{e}^{-2\kappa(\mathrm{T}-\mathrm{t})}-1}{2\kappa}\right)$$

따라서 위험 중립 측도에서 $\mathrm{P(t,T)}$는 정규분포의 적률생성함수(Moment Generate Function)에 따라 다음과 같다.

※ $\mathrm{E}(\mathrm{e}^{\mathrm{tx}}) = e^{\mu t+\frac{1}{2}(\sigma t)^2}$ MGF of Normal Distribution 따라서,

$$\mathrm{P(t,T)} = \hat{E}\left(\mathrm{e}^{-\int_{\mathrm{t}}^{\mathrm{T}}\alpha(u)du - x(t)B(t,T)+\frac{1}{2}V(t,T)}\right)$$

0시점 시장에서 관측된 만기 T인 Zero Bond $P^{\mathrm{M}}(0,T)$가 다음을 만족한다.

$$\mathrm{P}^{\mathrm{M}}(0,T) = \mathrm{e}^{-\int_0^{\mathrm{T}}\alpha(u)du+\frac{1}{2}V(0,T)}$$

$$\mathrm{e}^{-\int_0^{\mathrm{T}}\alpha(u)du} = \mathrm{P}^{\mathrm{M}}(0,T)\mathrm{e}^{-\frac{1}{2}V(0,T)}$$

따라서

$$\mathrm{e}^{-\int_{\mathrm{t}}^{\mathrm{T}}\alpha(u)du} = \frac{P^{\mathrm{M}}(0,T)}{P^{M}(0,t)}e^{-\frac{1}{2}(V(0,T)-V(0,t))}$$

$$\mathrm{P}_{\mathrm{HW}}^{x_i}(\mathrm{t},\mathrm{T}) = \frac{P^{\mathrm{M}}(0,T)}{P^{M}(0,t)}\exp\left(-x_iB(t,T)+\frac{1}{2}\left(V(t,T)-V(0,T)+V(0,t)\right)\right)$$

$$\mathrm{P}_{\mathrm{HW}}^{x_i}(\mathrm{t},\mathrm{T}) = \frac{P^{\mathrm{M}}(0,T)}{P^{M}(0,t)} \exp\left(-x_i \times B(t,T) + QVT(t,T)\right)$$

$$\mathbf{VT_{2F}(T_1, T_2, \kappa_1, \kappa_2, \sigma_1, \sigma_2)}$$
$$\approx \frac{\boldsymbol{\sigma_1\sigma_2}}{\kappa_1\kappa_2}\left(\mathrm{T}-\mathrm{t}+\frac{\mathrm{e}^{-\kappa_1(\mathrm{T}-\mathrm{t})}-1}{\kappa_1}+\frac{\mathrm{e}^{-\kappa_2(\mathrm{T}-\mathrm{t})}-1}{\kappa_2}-\frac{\mathrm{e}^{-(\kappa_1+\kappa_2)(\mathrm{T}-\mathrm{t})}-1}{(\kappa_1+\kappa_2)}\right)$$

$$\mathbf{CrossTerm_{2F}(t,T)} = \mathbf{2\rho\frac{1}{2}\left(VT_{2F}(t,T,\kappa_1,\kappa_2,\sigma_1,\sigma_2) - VT_{2F}(0,T,\kappa_1,\kappa_2,\sigma_1,\sigma_2) + VT_{2F}(0,t,\kappa_1,\kappa_2,\sigma_1,\sigma_2)\right)}$$

$$\mathrm{P}_{\mathrm{HW2F}}^{x_i y_i}(\mathrm{t},\mathrm{T}) = \frac{P^{\mathrm{M}}(0,T)}{P^{M}(0,t)} \exp\left(-x_i B_x(t,T) + QVT_x(t,T) - y_i B_y(t,T) + QVT_y(t,T) + \mathrm{CrossTerm}_{2\mathrm{F}}(\mathrm{t},\mathrm{T})\right)$$

· $\boldsymbol{B(t,T,\kappa)}$와 $\boldsymbol{VT(t,T,\kappa)}$은 다음과 같이 구현한다.

```
double B_s_to_t(
    double kappa,
    double s,
    double t
)
{
    return (1.0 - exp(-kappa * (t - s))) / kappa;
}
```

```
double V_t_T(
    double kappa,
    double kappa2,
    double t,
    double T,
    double vol,
    double vol2
)
{
    return vol * vol2 / (kappa + kappa2) * (T - t + (exp(-kappa * (T - t)) - 1.0) / kappa + (exp(-kappa2 * (T - t)) - 1.0) / kappa2 - (exp(-(kappa + kappa2) * (T - t)) - 1.0) / (kappa + kappa2));
}
```

5. **반복되는 연산을 피하기 위해 $\mathbf{XA}$, $\mathbf{XV}$, $\boldsymbol{B(t,T,\kappa)}$, $\boldsymbol{VT(t,T,\kappa,\{\sigma\})}$는 시뮬레이션 전에 미리 Generate 해놓고 epsilon을 시뮬레이션을 통해 산출하여 Short-Rate path 시뮬레이션한다.**

$$\mathbf{x}(\mathbf{t_{i+1}}) = \mathbf{XA}(\boldsymbol{t_i}, \boldsymbol{t_{i+1}}, \boldsymbol{\kappa}) \cdot \boldsymbol{x}(\boldsymbol{t_i}) + \boldsymbol{XV}(\boldsymbol{t_i}, \boldsymbol{t_{i+1}}, \boldsymbol{\kappa}, \{\boldsymbol{\sigma}\}) \cdot \boldsymbol{\epsilon_i}$$

$$\boldsymbol{P_{HW}(t,T)} = \mathrm{E}\left(\mathrm{e}^{-\int_t^T (x_u+\phi_u)du} | F_t\right) = \mathbf{e}^{-\boldsymbol{x_t} \cdot \boldsymbol{B(t,T,\kappa)} + \boldsymbol{QVT(t,T,\kappa)}}$$

여기서 $\mathbf{XA}, \mathbf{XV}$는 Short Rate의 시뮬레이션 t마다 dt 간격으로 생성

2D Matrix Size = (ShortRate시뮬레이션 커브개수, 시뮬레이션의 날짜개수)

또한 $\boldsymbol{B(t,T,\kappa)}$**와** $\frac{\mathbf{1}}{\mathbf{2}}\boldsymbol{VT(t,T,\kappa\{\sigma\})}$**는** 각 Rcv, Pay Leg의 기초금리의 만기까지 사전에 Generate해야함.

$\boldsymbol{B_{Rcv}(t_{Simul}, T_{Rcv}, \kappa)}$, $\boldsymbol{B_{Pay}(t_{Simul}, T_{Pay}, \kappa)}$ **사이즈와**

$\boldsymbol{VT_{Rcv}(t_{Simul}, T_{Rcv}, \kappa, \{\sigma\}), VT_{Pay}(t_{Simul}, T_{Pay}, \kappa, \{\sigma\})}$ **사이즈는**

3D Matrix Size =

(ShortRate시뮬레이션 커브개수, 시뮬레이션의 날짜개수, Leg별 쿠폰개수)

6. **이후 기초금리를 계산한다.**

$$\mathrm{R_{HW}(t,T)} = \frac{1 - P_{HW}(t, T_N)}{\sum_1^N [\Delta T_i \times P_{HW}(t, T_i)]}$$

예를 들어 2년 만기 스왑금리를 기초금리라고 가정한다면,

1년 뒤의 시뮬레이션된 금리 산출과정은 다음과 같다.

$$\mathrm{P_{HW}}(1,1.25) = \frac{P(0,1.25)}{P(0,1)} \exp\left(-r_1 \times B(1,1.25) + QVT(1,1.25)\right)$$

$$\mathrm{P_{HW}}(1,1.5) = \frac{P(0,1.5)}{P(0,1)} \exp\left(-r_1 \times B(1,1.5) + QVT(1,1.5)\right)$$

$$\mathrm{P_{HW}}(1,1.75) = \frac{P(0,1.75)}{P(0,1)} \exp\left(-r_1 \times B(1,1.75) + QVT(1,1.75)\right)$$

$$\mathrm{P_{HW}}(1,2) = \frac{P(0,2)}{P(0,1)} \exp\left(-r_1 \times B(1,2) + QVT(1,2)\right)$$

$$\mathrm{P_{HW}}(1,2.25) = \frac{P(0,2.25)}{P(0,1)} \exp\big(-r_1 \times B(1,2.25) + QVT(1,2.25)\big)$$

$$\mathrm{P_{HW}}(1,2.5) = \frac{P(0,2.5)}{P(0,1)} \exp\big(-r_1 \times B(1,2.5) + QVT(1,2.5)\big)$$

$$\mathrm{P_{HW}}(1,2.75) = \frac{P(0,2.75)}{P(0,1)} \exp\big(-r_1 \times B(1,2.75) + QVT(1,2.75)\big)$$

$$\mathrm{P_{HW}}(1,3) = \frac{P(0,3)}{P(0,1)} \exp\big(-r_1 \times B(1,3) + QVT(1,3)\big)$$

$$\mathrm{R_{HW}^{Swap}}(1,3) = \frac{1 - P_{HW}(1,3)}{\sum_1^8 [\Delta T_i \times P_{HW}(1, T_i)]}$$

끝.